WHEN TIME-OF-USE COSTS TOO MUCH

DAWN BLASCO

(left to right) Ross Butcher, Frank Swatek, and author Chris Staskewicz have finished installing a sharp SRS solar system on cement/composite roofing tiles.

The California Solar Initiative (CSI) has run into some unexpected hurdles for certain residential customers planning to install PV modules on their homes. Under the terms for those requesting rebates, households are required to switch from their standard billing schedule to a time-of-use (TOU) schedule. In certain cases, a household will experience higher costs and reduced savings by making this switch.

For Southern California Edison (SCE) customers, TOU was designed to reduce electricity costs for those who consume most of their electricity during off-peak hours (before 10 am and after 6 pm) by charging them less per kilowatt hour (kWh) (see Table 1). Before the impact of TOU rates on the CSI program caught the attention of the media, the solar industry, and Governor Schwarzenegger, select households had subscribed to it voluntarily in order to reduce their electric bills. For example, families with children at school and parents working outside the home rely more on off-peak energy during the early morning and late evening than on-peak energy during the day. By paying lower rates for off-peak electricity, these families will pay less on a TOU schedule than they will on the standard residential schedule. Other investor-owned and municipal utility companies in California have similar TOU programs.

Why Time-Of-Use Doesn't Always Pay

Higher costs under TOU result when a household consumes most of its electricity during on-peak hours—that is, between 10 am and 6 pm, for SCE customers. Additional rate imbalances occur in June through September, when rates are higher due to increased demand for electricity (primarily air conditioning). As of March 2007, a typical household serviced under the TOU Domestic Level 2 (D-2) rate schedule could expect to pay summer rates of approximately \$0.30/kWh during on-peak hours versus \$0.16/kWh during off-peak hours.

In order to measure the effects of switching to TOU, either by electing into the program or by installing solar under the CSI rebate program, the company I cofounded, Progressive Power Group, Incorporated, in Long Beach, California, wanted to compare billing under TOU with the household's current residential billing schedule. In most cases, a single-family residence, condominium, or apartment served by SCE qualifies for the basic rate under Schedule D (see Table 2). Under this arrangement, households are allotted a baseline electricity usage per day, and any consumption over the baseline is billed at a higher rate, as shown in the table.

While most regions have similar baseline allocations during the winter, the city of Palm Desert and other Inland Empire communities have baseline allocations up to 3.3 times greater during the summer than those of any other region serviced by SCE. Consequently, households in these regions receive substantial discounts under Schedule D, regardless of the time of day. It is essential to understand this aspect of the billing structure when you advise clients about solar-electric systems and other energy-saving measures.

In Palm Desert and the surrounding communities, people use a great deal of electricity during peak hours. A survey of residential clients conducted by our firm found that usage averages about 1,000 kWh per month during the winter, and about 2,250 kWh per month during the summer. Furthermore, most customers surveyed said that 75% or more of their consumption occurs during on-peak hours, regardless of the season. This includes refrigeration, air conditioning, and pool equipment. The majority of households surveyed included at least one stay-at-home family member, another reason for heavy on-peak consumption.

This analysis demonstrates that, even when the same amount of electricity is consumed under the two schedules, annualized costs under TOU-D-2 are greater than annualized costs under Schedule D, due to premium on-peak charges and the skewed baseline allocation during the summer (see Table 3). It is also worth noting that winter-related costs are less under TOU than under Schedule D. This is consistent with guidelines established by SCE.

A cost comparison under the two schedules makes clear the economic consequences of installing solar electricity on a residential site. The following example illustrates the adjusted electricity costs a household would pay under the two schedules if it

Table 1. SCE Time-of-Use

Summer—On peak	$0.30
Summer—Off peak	$0.16
Winter—On peak	$0.18
Winter—Off peak	$0.16

Note: Average $/kWh excludes customer fees and taxes.

Table 2. SCE Schedule D

Tier 1—Baseline	$0.12
Tier 2—101–130%	$0.14
Tier 3—131%–200%	$0.23
Tier 4—201%–300%	$0.26
Tier 5—Over 200%	$0.30

Note: Average $/kWh excludes customer fees and taxes.

Table 3. Cost Comparisons

	kWh	Sch D	TOU[1]
Summer	2,250	$ 332	$ 599
Winter	1,000	$ 204	$ 178
Annual	17,000	$2,960	$3,816

[1] 75% summer on peak, 75% winter on peak related charges.

Table 4. Adjusted Electricity Costs [1]

	kWh	Sch D	TOU
Summer	1,814	$ 233	$ 498
Winter	624	$ 102	$ 113
Annual	12,249	$1,744	$2,891

[1] Includes offset from a 2.6 kW system

Table 5.

		Annual savings		Savings per $ invested	
Modules[1]	Net cost[2]	Sch D	TOU	Sch D	TOU
6	$ 5,069	$ 539	-$460	0.106	-0.091
10	$ 9,781	$ 877	-$195	0.090	-0.020
14	$14,493	$1,216	$ 69	0.084	0.005
18	$19,206	$1,531	$333	0.080	0.017
22	$23,918	$1,794	$598	0.075	0.025
26	$28,631	$1,995	$862	0.070	0.030

[1] 187 Watt modules
[2] After CSI rebate of $2.50 per Watt (88% of STC) and $2,000 federal tax credit

installed a 2.6 kW solar system consisting of 14 Sharp 187W modules in Palm Desert (see Table 4). The approximate net cost for such a system, after a federal tax credit of $2,000 and a CSI rebate of $5,700 ($2.50/W with a wattage conversion factor of 88%, reflecting real-world operation) is $14,500.

When we compare the adjusted costs with the original Schedule D bill, we find that TOU clearly underperforms—with a savings of approximately $70 versus $1,216 under Schedule D. That is, for every net dollar invested, the solar system will return 8¢ in annual savings under Schedule D, but less than 1¢ under TOU. Moreover, a household that installed a system smaller than 2.6 kW could experience an increase of up to $460 in annual costs (see Table 5).

Petitioning the Powers That Be

A petition submitted before the California Public Utility Commission (CPUC) claimed that "customers are forgoing the purchase of a solar energy system due to the negative financial impacts from TOU rate structures. They claim this is directly at odds with the goal of [CSI] to create the maximum incentive for customers to install solar."

Since the CSI program began, in January 2007, there has been a lot of debate about requiring households to switch over to TOU schedules upon installing solar and claiming the cash rebate. The underlying consensus is that some people benefit from doing so, while others do not. Currently, many homeowners in Palm Desert and elsewhere are postponing solar installations until their rate schedules can be changed, to create a balanced economic benefit for everyone. If the state of California wants to create a positive environment for solar and other green initiatives, the legislature must give consumers a way to maximize the economic returns of going solar. After all, solar energy for the home is expensive, and households will be slow to switch to solar until regulations make it worth their money to do so.

Ironically, households that burden the grid with on-peak demand will continue to do so whether they install solar or not. If they do not install solar, they will continue to be billed at Schedule D rates, which are independent of time-of-use and are subsidized through higher baseline energy allocations. That is, Schedule D penalizes for incremental consumption only above a certain threshold, whether it's morning, noon, or night. Since such a household would never switch to solar on a TOU schedule, giving it the option to remain on Schedule D will give it an incentive to go solar. Doing so will offset most of its on-peak demand, which in turn will reduce the demand on the grid during on-peak hours. From both the household's and the grid's perspective, this is a winning solution.

Last-Minute Update

The day before I submitted this article for publication, the CPUC voted to modify TOU billing schedules for households and businesses that install solar and receive rebates under the CSI program. In the agenda, the CPUC recognized the "potential for adverse rate impacts for some portion of customers who install solar energy systems."

That's not the end of the story, however. The CPUC granted a temporary stay on switching to TOU schedules until a comprehensive TOU structure can be developed to maximize the economic benefit for all solar participants. But until this hypothetical schedule can be drafted, households should be jubilant. The stay requires Pacific Gas and Electric Company, SCE, and San Diego Gas and Electric Company to identify and inform households that installed solar under the CSI program that they are eligible to switch to their older schedule (Schedule D) or to fixed-rate schedules (General Services for commercial properties). In addition, these households are to receive credits for costs that they incurred during the past five months as a result of being on the TOU schedule.

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